A Additional Evaluation Details

A.1 Diversity of Reasoning Models

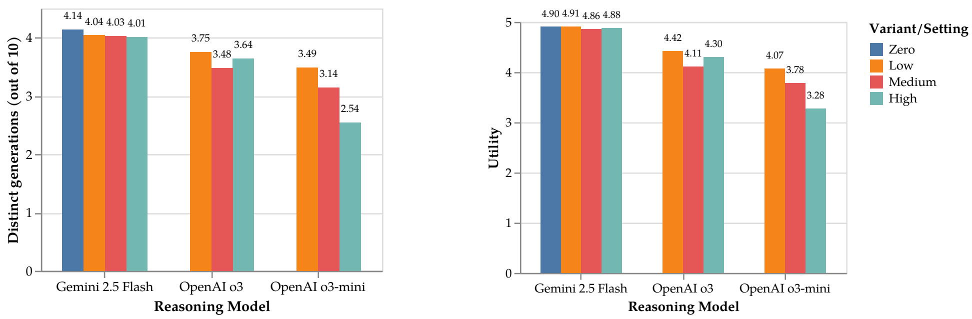

Figure 7: Diversity and utility performance of reasoning models on NoveltyBench. OpenAI's models demonstrate a trend where increased reasoning effort leads to reduced diversity and utility.

Our core analysis focuses on the 20 most representative (non-reasoning) language models from major providers to establish a comprehensive baseline across model families and sizes, since reasoning capabilities are not yet widely available across model families at the time of writing. That said, as more frontier models carry out an explicit reasoning process before responding to user queries, we believe it would be interesting to see how such reasoning affects diversity.

To this end, we evaluate OpenAI and Google's latest reasoning models (o3, o3-mini, and Gemini-2.5-flash) for which inference-time reasoning effort can be explicitly set, and explored how they impact diversity and utility on NoveltyBench. For the Gemini model, setting the reasoning budget had minimal effect on our evaluation. On the other hand, we see a trend with OpenAI's models where additional reasoning seems to lead to less diverse and less useful outputs. We note that although reasoning models are usually used to solve problems with a single correct answer (e.g., math and coding), preserving diversity is still critical for the effectiveness of inference-time scaling strategies such as pass@k. Future work should further investigate how generation diversity affects creativity in problem solving, and its implications on inference-time scaling.

A.2 Prompt for Selecting Diverse Prompts

The following prompt was used with GPT-4o to select prompts from the WildChat dataset that allow for diverse responses:

A.3 Example Prompts from NB-WILDCHAT

The following table presents a sample of prompts from the NB-WILDCHAT dataset that were selected using the GPT-4o classifier:

A.4 Evaluation of equivalence classification methods

To determine the equivalence of two language model generations, we used a variety of existing text similarity metrics such as BERTScore, BLEU, ROUGE, as well as fine-tuning OpenAI models (`GPT-3.5-turbo` and `GPT-4o-mini`) and a pre-trained DeBERTa model (`microsoft/deberta-v3-large`) on a manually labeled dataset of 1,000 pairs of LM-generated text.

On a validation set of 100 pairs, we found that a DeBERTa reached the highest AUC score of 0.81, outperforming both fine-tuned OpenAI models and heuristic-based methods

```
You are helping select prompts for a benchmark that measures language models'
ability to generate diverse, high-quality alternative answers. For a prompt
to qualify, it should:
1. Allow diverse responses: The prompt should enable multiple valid distinct
responses. For example, a prompt that asks for a salmon recipe, a chess
move in a given position, or a continuation of a story would allow diverse
responses. In contrast, a prompt that asks for a specific fact, or a rewrite
of input text would not.
2. Both the prompt, and the desired response should be in English.
3. Request a natural language response. Requests that ask for code, images
or other output formats do not qualify.
4. Make a single clearly interpretable request. For example, "recommend a
reliable espresso machine" is clear, while "espresso machine" is not.

Classify the following prompt based on these criteria, and format the provided
prompt. Output using the provided JSON format.
```

Table 3: System prompt used for selecting diverse prompts from the WildChat dataset

Example Prompts
Describe me as a smart, funny, and kind guy who could change anybody's life for an "about me" section on a dating app. Make it professional and smart.
Write an example illustrating how climate change can be considered a natural phenomenon rather than man-made.
How can I make a collage more interesting?
Value yourself as a star, sun, and a tall man.
Can you introduce "audit" as a research method in social science?
Write a story about a wizard being sent back in time to when he was born after dying before mastering all types of magic.
Write a background of study on the topic "Navigating the Role of Campus Directors as Managers: A Narrative Inquiry of Their Lived Experiences."
How can I learn to code?
Write a story about a man who uses a magical remote to turn himself into his dream girlfriend, changing himself one step at a time.
What are the best free VPNs? List the reasons and reviews.

Table 4: Randomly sampled prompts from the NB-WILDCHAT dataset

significantly. After choosing an appropriate confidence interval, the fine-tuned DeBERTa model has an accuracy of 71.0% and an F1 score of 0.811.

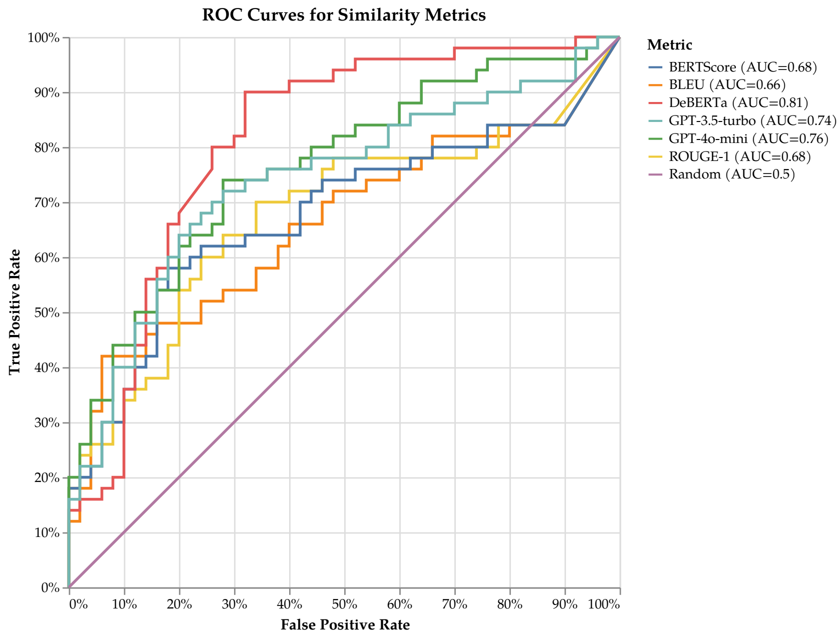

Figure 8: ROC curve for various similarity classification methods.

A.5 Human Annotations

Eight human annotators, drawn from members of our institution, responded to each prompt in NB-CURATED. Examples of the human annotations corresponding to the prompts from Table 2 are presented in Table 5.

Suggest a name for a dappled-gray filly living in the mountains.	**Name one person who won the Fields Medal.**	**Roll a make-believe 20-sided die.**	**What's the best car to get in 2023?**
Jumpy	Kunihiko Kodaira	17	Subaru BRZ
Maximus	Akshay Venkatesh	18	Honda Civic
Oolong	James Maynard	9	Toyota Camry
Winter	Gerd Faltings	20	2025 Rivian R1S
Teazle	Maryam Mirzakhani	7	Prius
Tony	Maryna Viazovska	20	We decided on a Mazda CX5. It was the best value per dollar.
Monte	June Huh	19	Tesla Model Y
Greg	Maryam Mirzakhani	13	Volvo XC90

Table 5: Human responses to prompts from NB-CURATED.